Lesson 02

The basis representation of integers

Content

Objectives

At the end of this session you should be able to

- Explain the basis representation
- Develop the divisibility criteria of some integers
- use the basis representation in elementary level applications

2.1 Introduction

The invention of numeration system is one of the greatest inventions of humanity. Among the earliest civilizations Babylonians, Romans and Indians use different numeration systems.

The numeration system in commonly use in today is the Hindu Arabic numeration system which is originated from early Indian civilization. In Hindu Arabic numeration system there are ten symbols, namely 0,1,2,3,4,5,6,7,8,9. The selection of ten digits might be having 10 fingers for the humans. Any positive integer can be represented by using only the ten symbols.

In the decimal system the value of a digit depends on its place in the number. Each place has a value of 10 times the place to the right. As an example three hundred eight is written 308, which stands for $3 \times 10^2 + 0 \times 10^1 + 8 \times 10^0$.

2.2 The basis representation

In the introduction we mentioned that, in Hindu Arabic numeration system, base 10 representation adapted because of the 10 fingers in human hands. Other than this reason the base 10 has no any exception. Instead of 10, if we choose any integer b , which is greater than 1, then everything is working properly as in base 10 case.

Any integer N can be represent as, $N = a_m b^m + a_{m-1} b^{m-1} + \cdots + a_2 b^2 + a_1 b^1 + a_0$, where $a_i \in \{0, 1, 2 \ldots (b-1)\}$. We use the shorter notation $(a_m a_{m-1} \ldots a_2 a_1 a_0)_b$ to identify the base b representation of this number. Usually in the base 10 representation we don't write the subscript 10.

2.2.1 Example

Find the base 10 representation of $(453)_7$

Answer: $(456)_7 = 4 \times 7^2 + 5 \times 7^1 + 3 \times 7^0 = 196 + 35 + 3 = 234 = (234)_{10}$. Hence the base 10 representation of $(453)_7$ is 234.

2.2.2 Example

Find the base 2 representation of 214 {i.e. $(214)_{10}$}.

Answer: $214 = 1 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$. Hence the base 2 representation of 214 is $(11010110)_2$.

2.2.3 Theorem - Basis representation

Any integer N can be written uniquely in terms of powers of b as,

$N = a_m b^m + a_{m-1} b^{m-1} + \cdots + a_2 b^2 + a_1 b + a_0$, where $a_i \in \{0, 1, 2 \ldots (b-1)\}$, $b > 1$ and $b \in \mathbb{N}$.

Proof:- Later (After learning Division Algorithm)

2.3 Divisibility criteria for base 10 representation

Recall the several divisibility criteria you learned in your middle school. Divisibility criteria is a shorter method to check whether a positive integer is divided by a fixed positive integer. For example :

- The ones place digit of a number is divisible by 2 if and only if that number is divisible by 2.

- The ones place digit of a number is either 0 or 5 if and only if the number is divisible by 5.

- The ones place digit of a number is 0, if and only if the number is divisible by 10.

To find the criteria for divisible by 9 and 3, the concept of digital root of a number is important.

2.3.1 Definition

The digital root of a number is calculated by adding up all the digits of that number (and adding the digits of the sums if necessary), until a single digit is defined as the digital root of the relevant number.

2.3.2 Example

Find the digital root of the number 9887 .

Answer: 9+8+8+7=32 and 3+2=5. Hence the digital root of 9887 is 5.

2.3.3 Theorem

If the base 10 representation of a number is $a_m a_{m-1} \dots a_2 a_1 a_0$, then 9 divides $a_m a_{m-1} \dots a_2 a_1 a_0$ if and only if 9 divides $a_m + a_{m-1} + \cdots a_2 + a_1 + a_0$.

Proof:

Observe that $a_m a_{m-1} \dots a_2 a_1 a_0 = 10^m a_m + 10^{m-1} a_{m-1} + \dots + 10^2 a_2 + 10 a_1 + a_0$.

Hence,

$$a_m a_{m-1} \dots a_2 a_1 a_0 = (10^m - 1) a_m + (10^{m-1} - 1) a_{m-1} + \dots + (10^2 - 1) a_2 + (10 - 1) a_1$$

$$+ a_m + a_{m-1} + \cdots a_2 + a_1 + a_0 .$$

But 9 divides $(10^m - 1)$ for every $m \in \mathbb{N}$.

Hence from above equation 9 divides $a_m a_{m-1} \dots a_2 a_1 a_0$ if and only if 9 divides $a_m + a_{m-1} + \cdots a_2 + a_1 + a_0$.

Note: From above theorem and the concept of the digital root, it is clear that, the number is divisible by 9 if and only if it's digital root is divisible by 9.

Activity 1

Prove that If the base 10 representation of a number is $a_m a_{m-1} \dots\ a_2 a_1 a_0$, then 3 divides $a_m a_{m-1} \dots\ a_2 a_1 a_0$ if and only if 3 divides $a_m + a_{m-1} + \cdots\ a_2 + a_1 + a_0$.

The following theorem is useful to establish divisibility criteria of various integers.

2.3.4 Theorem

Let $N = a_m a_{m-1} \dots\ a_2 a_1 a_0 = a_m 10^m + \cdots + a_2 10^2 + a_1 10 + a_0$ be the decimal expansion of of the positive integer N and let $T = a_0 - a_1 + a_2 - \dots + (-1)^m a_m$. Then 11 divides N if and only if 11 divides T

Proof:

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2.4 Practical applications

Modular arithmetic and basis representation has many practical applications. We present here a very few applications.

2.4.1 Check digit

Suppose a check or money order has 10 digits identification number. Then it can be added another digit after those 10 digits. The value of this last digit should be the remainder when divides first 10 digits number by 9. This 11th digit is called the check digit.

For example, if 3953988164 is the 10 digits identification number, then check digit should be 3953988164 check digit is 2. Now with this check digit, the new identification number is 39539881642. If the number 39539881642 were incorrectly entered in to a computer (programmed one to identify the check digit) say, as 39559881642 (an error in the fourth position) then the machine calculates the check digit and error will be detected.

Review Questions

(i) Prove that $10^i - 4$ $divided\ by$ 6 for every positive integer i.(Hint: use induction)
Deduce that 6 divides N if and only 6 divides $a_0 + 4a_1 + 4a_2 + \cdots + 4a_m$

(ii) Write down any three-digit number. Again, write down same number right to the last digit of first three-digit number. Now you have 6-digit number. Divides that number by 7. Now divides the answer by 11. Again, divides that answer by 13.

Wait a minute. Can you see that you can divide that 6 digit number by 7, 11 and 13 without any remainder? Why this is happening

Grade 4+

අණ්ඩර අන්දර 19 - අරාබි නිසොල්ලාසය

මතකද ඔබට අරාබි නිසොල්ලාසය,
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එක් දහස් එක් රැයක්,
ඕවි හරියටම එක්
දහස් එක් රැයක්
පුරා, කතන්දර කියූ
ඒ කතාව.

ලියන්න, ඉලක්කම් තුනක සංඛ්‍යාවක්. නැවත එම සංඛ්‍යාවේ අග ඉලක්කමට දකුණු පසින් එම සංඛ්‍යාවම ලියන්න. ඉලක්කම් හයක සංඛ්‍යාවක් ඔබට දැන් ලැබී ඇත.

බෙදන්න, එම සංඛ්‍යාව හතෙන්. බෙදන්න ලැබෙන පිළිතුර එකොළහෙන්. බෙදන්න නැවතත් ලැබෙන පිළිතුර දහතුනෙන්.

මඳක් නවතින්න, පුදුමයක් නොවේද මෙය? දුටුවාද? හැම විටම ඉතිරි නොවී බෙදෙනා සැටි. කෙලෙසද මෙය සිදුවන්නේ?

මතකද ඔබට අරාබි නිසොල්ලාසය, තරුණියක් රජු හට එක් දහස් එක් රැයක්, ඕවි හරියටම එක් දහස් එක් රැයක් පුරා ...